R Console

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```
>
> library(ggplot2)
>
> new_window <- function() {
+   if (.Platform$OS.type == "windows") {
+     windows()
+   } else {
+     quartz()
+   }
+ }
>
> # ----------------------------
> # Q1a: Histogram - Math Score
> # ----------------------------
> new_window()
> hist(student_data$Math_Score,
+      col = "skyblue",
+      main = "Histogram of Math Scores",
+      xlab = "Math Score",
+      ylab = "Frequency",
+      border = "black",
+      breaks = 5)
>
> # ----------------------------
> # Q1b: Boxplot - Science Score by Gender
> # ----------------------------
> new_window()
> boxplot(Science_Score ~ Gender,
+         data = student_data,
+         col = c("pink", "lightblue"),
+         main = "Boxplot of Science Score by Gender",
+         xlab = "Gender",
+         ylab = "Science Score",
+         border = "black")
>
> # ----------------------------
> # Q2: Scatter Plot - Study Hours vs Math Score (colored by Gender)
> # ----------------------------
> new_window()
> print(
+   ggplot(student_data, aes(x = Study_Hours, y = Math_Score, color = Gender)) +
+     geom_point(size = 4) +
+     geom_smooth(method = "lm", se = FALSE, aes(group = 1), color = "black",
+                 linetype = "dashed") +
+     scale_color_manual(values = c("Female" = "red", "Male" = "blue")) +
+     labs(title = "Study Hours vs Math Score by Gender",
+          x = "Study Hours",
+          y = "Math Score",
+          color = "Gender") +
+     theme_minimal()
+ )
`geom_smooth()` using formula = 'y ~ x'
>
> # ----------------------------
```

R Console

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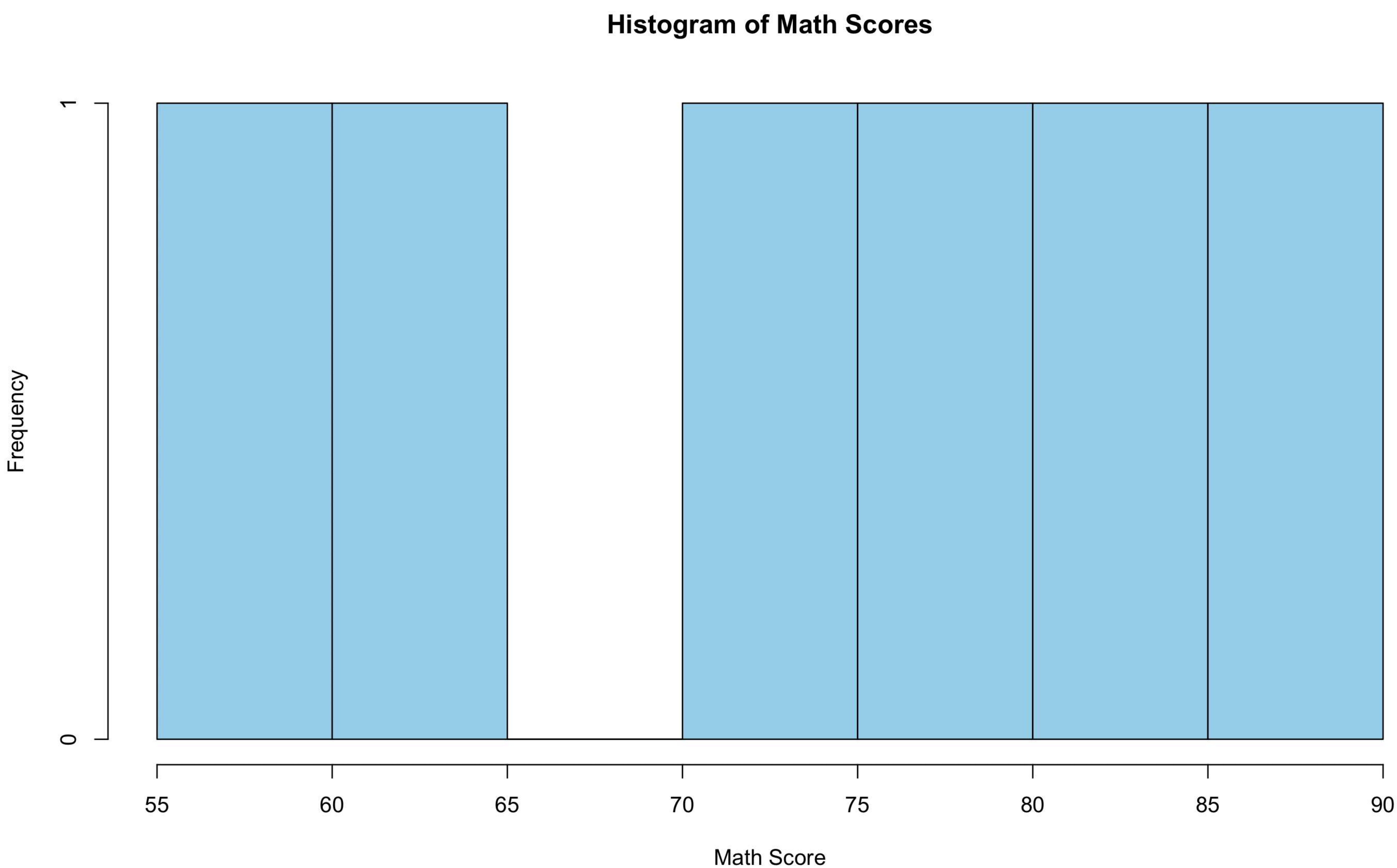
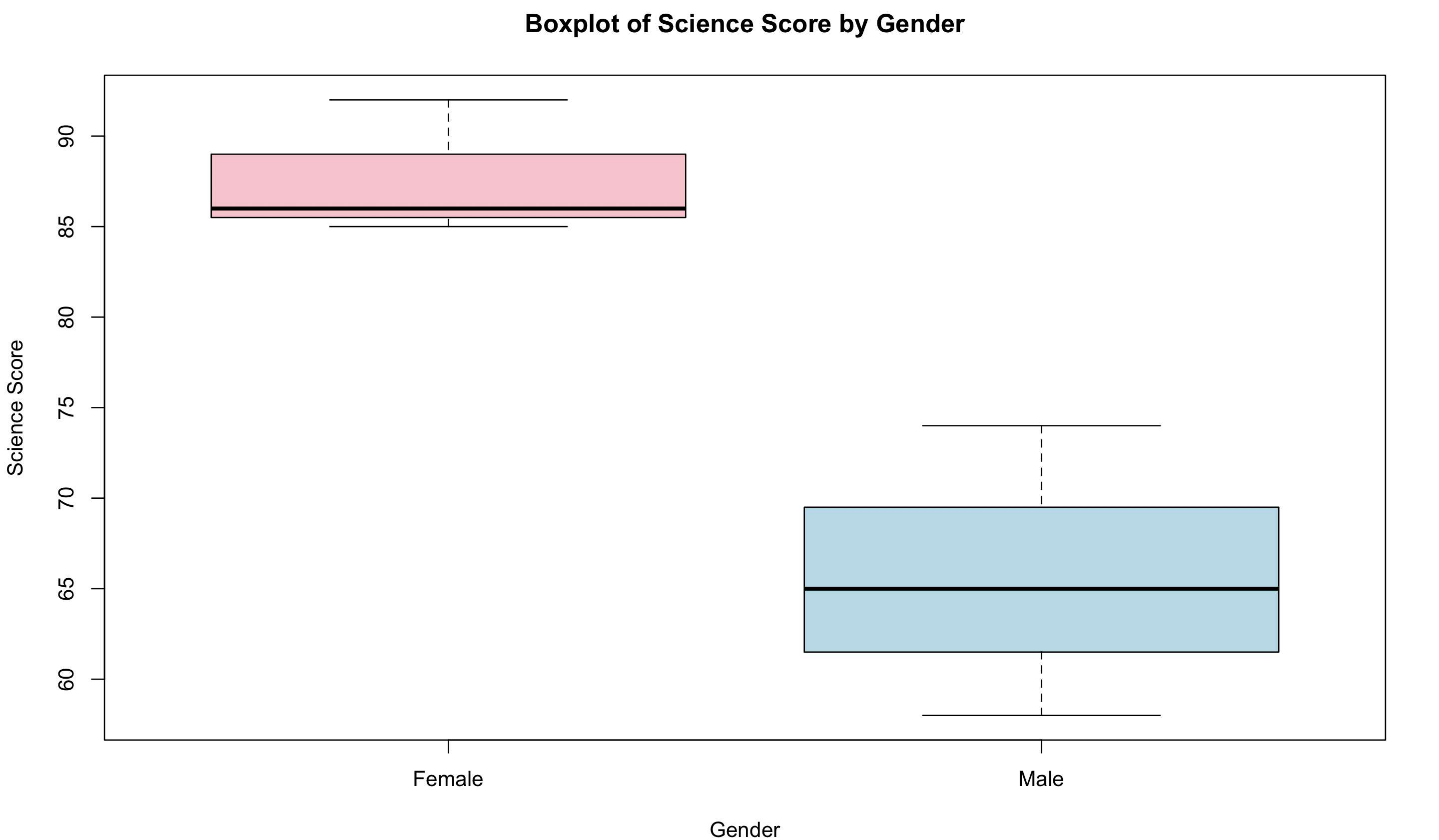
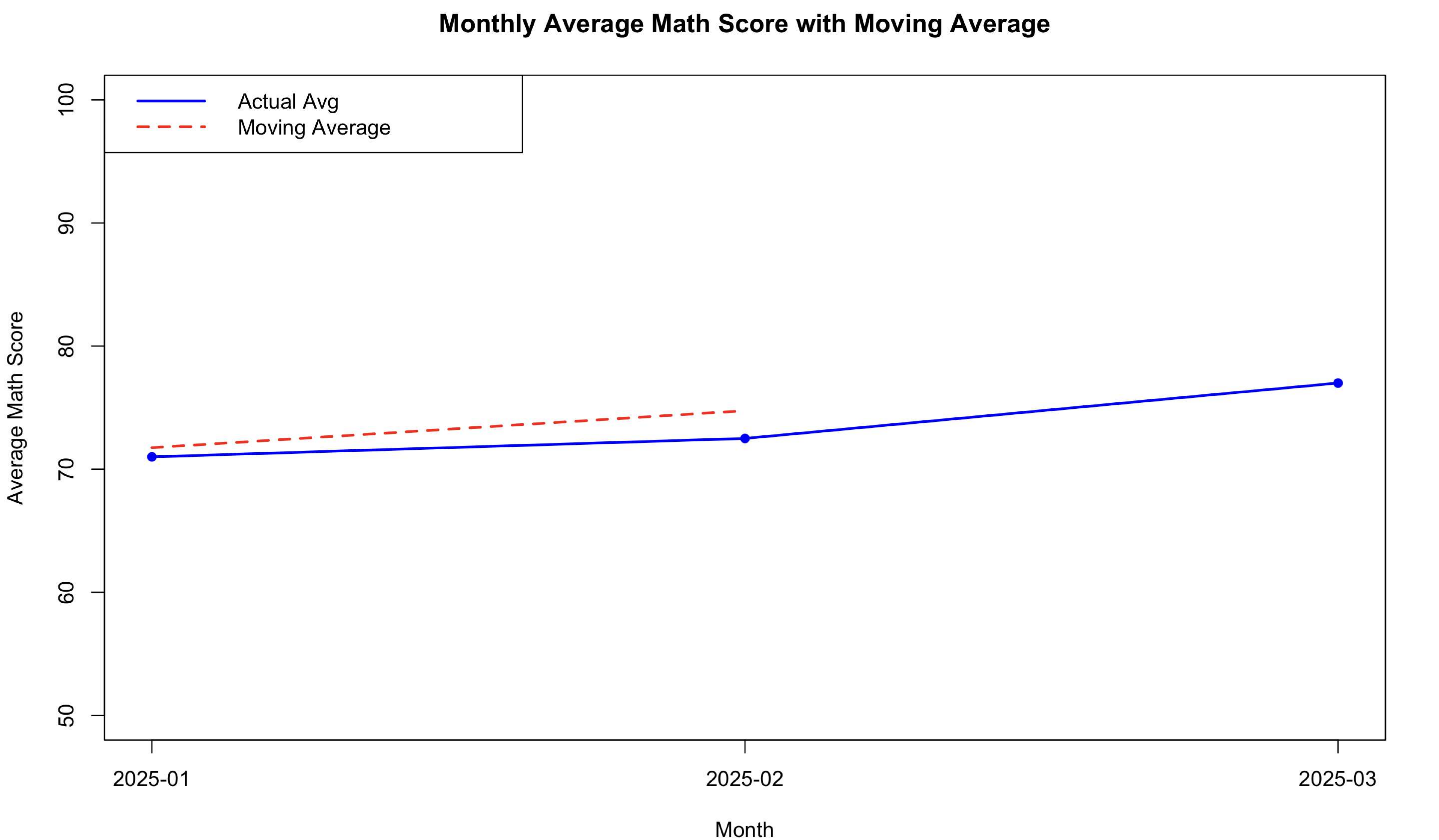
Help Search

```
+     geom_smooth(method = "lm", se = FALSE, aes(group = 1), color = "black",
+                 linetype = "dashed") +
+     scale_color_manual(values = c("Female" = "red", "Male" = "blue")) +
+     labs(title = "Study Hours vs Math Score by Gender",
+          x = "Study Hours",
+          y = "Math Score",
+          color = "Gender") +
+     theme_minimal()
+ )
`geom_smooth()` using formula = 'y ~ x'
>
> # -----------------------------
> # Q3: Date Conversion + Monthly Avg Math Score + Line Chart + Moving Average
> # -----------------------------
> student_data$Exam_Date <- as.Date(student_data$Exam_Date)
> student_data$Month <- format(student_data$Exam_Date, "%Y-%m")
>
> # Monthly average Math Score
> monthly_avg <- aggregate(Math_Score ~ Month, data = student_data, FUN = mean)
> monthly_avg <- monthly_avg[order(monthly_avg$Month), ]
> monthly_avg$Month_Index <- 1:nrow(monthly_avg)
>
> print(monthly_avg)
    Month Math_Score Month_Index
1 2025-01       71.0           1
2 2025-02       72.5           2
3 2025-03       77.0           3
>
> # Simple Moving Average function
> moving_avg <- function(x, n = 2) {
+   stats::filter(x, rep(1/n, n), sides = 2)
+ }
>
> monthly_avg$MA <- as.numeric(moving_avg(monthly_avg$Math_Score, n = 2))
>
> new_window()
> plot(monthly_avg$Month_Index, monthly_avg$Math_Score,
+      type = "o",
+      col = "blue",
+      lwd = 2,
+      pch = 16,
+      main = "Monthly Average Math Score with Moving Average",
+      xlab = "Month",
+      ylab = "Average Math Score",
+      xaxt = "n",
+      ylim = c(50, 100))
> axis(1, at = monthly_avg$Month_Index, labels = monthly_avg$Month)
> lines(monthly_avg$Month_Index, monthly_avg$MA,
+       col = "red", lwd = 2, lty = 2)
> legend("topleft",
+        legend = c("Actual Avg", "Moving Average"),
+        col = c("blue", "red"),
+        lty = c(1, 2), lwd = 2)
>
```

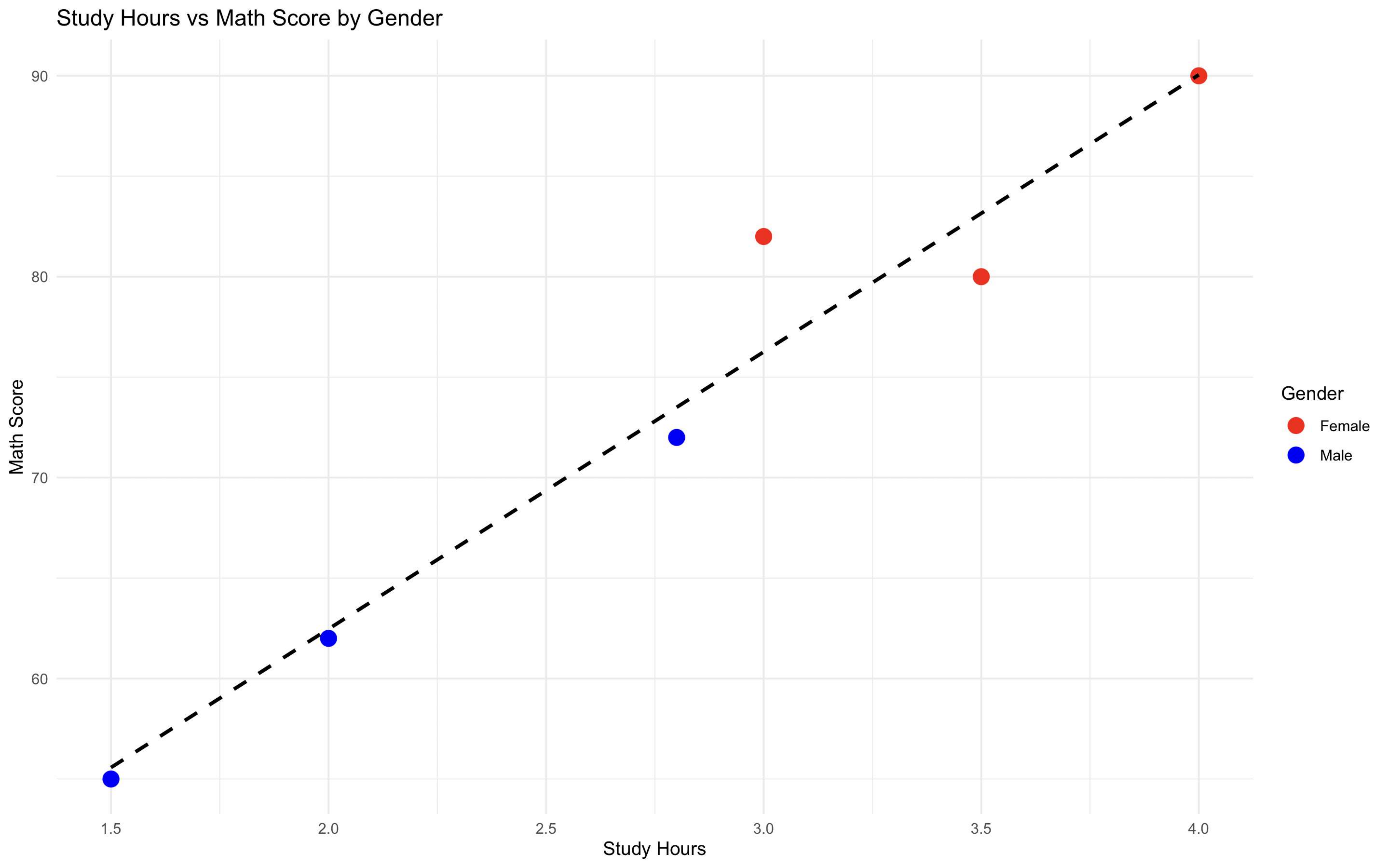

Histogram of Math Scores
Frequency
1
0
55
60
65
70
75
80
85
90
Math Score

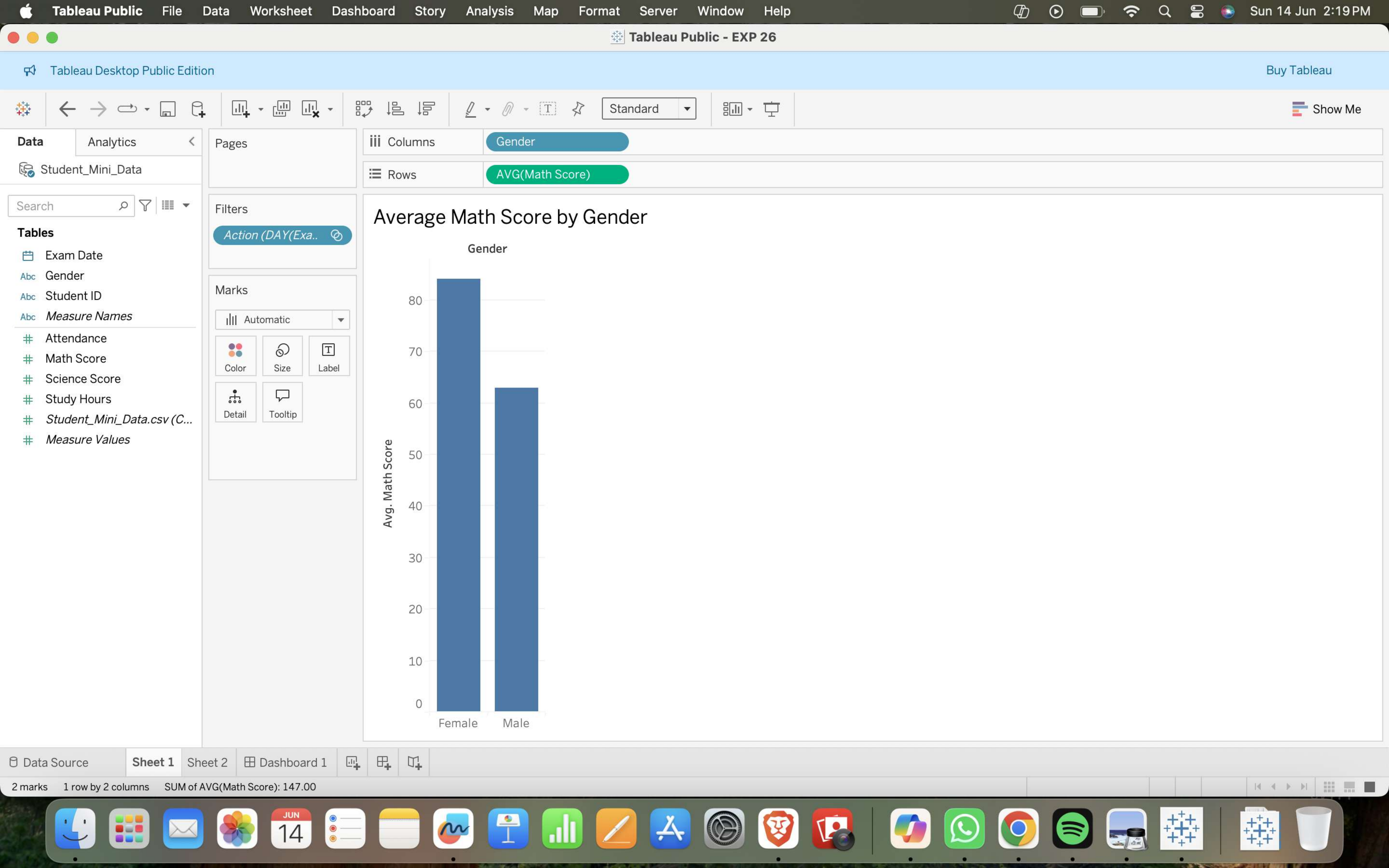


Boxplot of Science Score by Gender
Science Score
90
85
80
75
70
65
60
Female
Male
Gender



Monthly Average Math Score with Moving Average
Actual Avg
Moving Average
Average Math Score
100
90
80
70
60
50
2025-01
2025-02
2025-03
Month


Study Hours vs Math Score by Gender
90
80
70
60
Math Score
1.5
2.0
2.5
3.0
3.5
4.0
Study Hours
Gender
Female
Male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Student_Mini_Data
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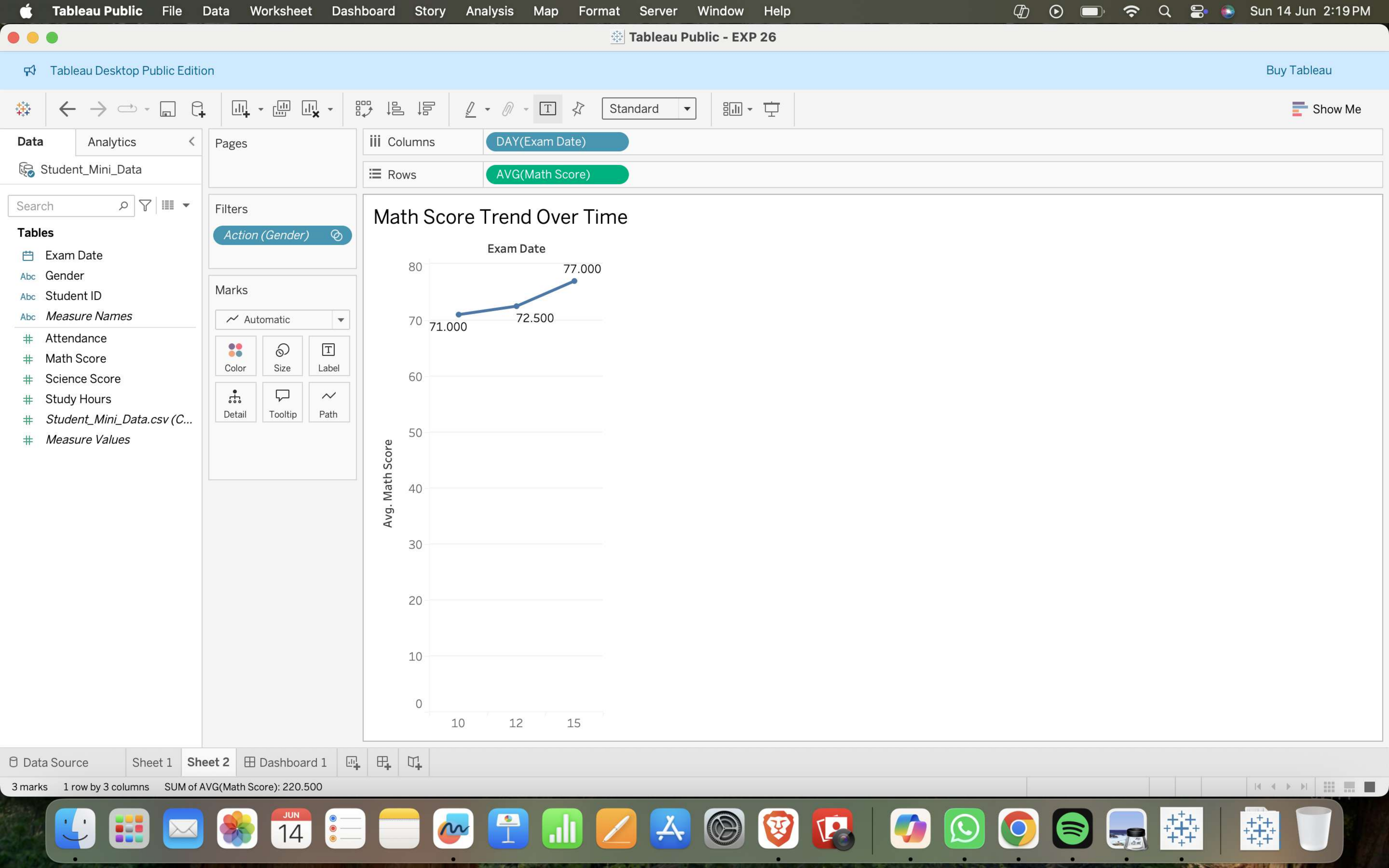
Tables
Exam Date
Gender
Student ID
Measure Names
Attendance
Math Score
Science Score
Study Hours
Student_Mini_Data.csv (C...
Measure Values
Pages
Filters
Action (DAY(Exa..
Marks
Automatic
Color
Size
Label
Detail
Tooltip
Columns
Gender
Rows
AVG(Math Score)
Average Math Score by Gender
Gender
Avg. Math Score
80
70
60
50
40
30
20
10
0
Female
Male
Data Source
Sheet 1
Sheet 2
Dashboard 1
2 marks
1 row by 2 columns
SUM of AVG(Math Score): 147.00



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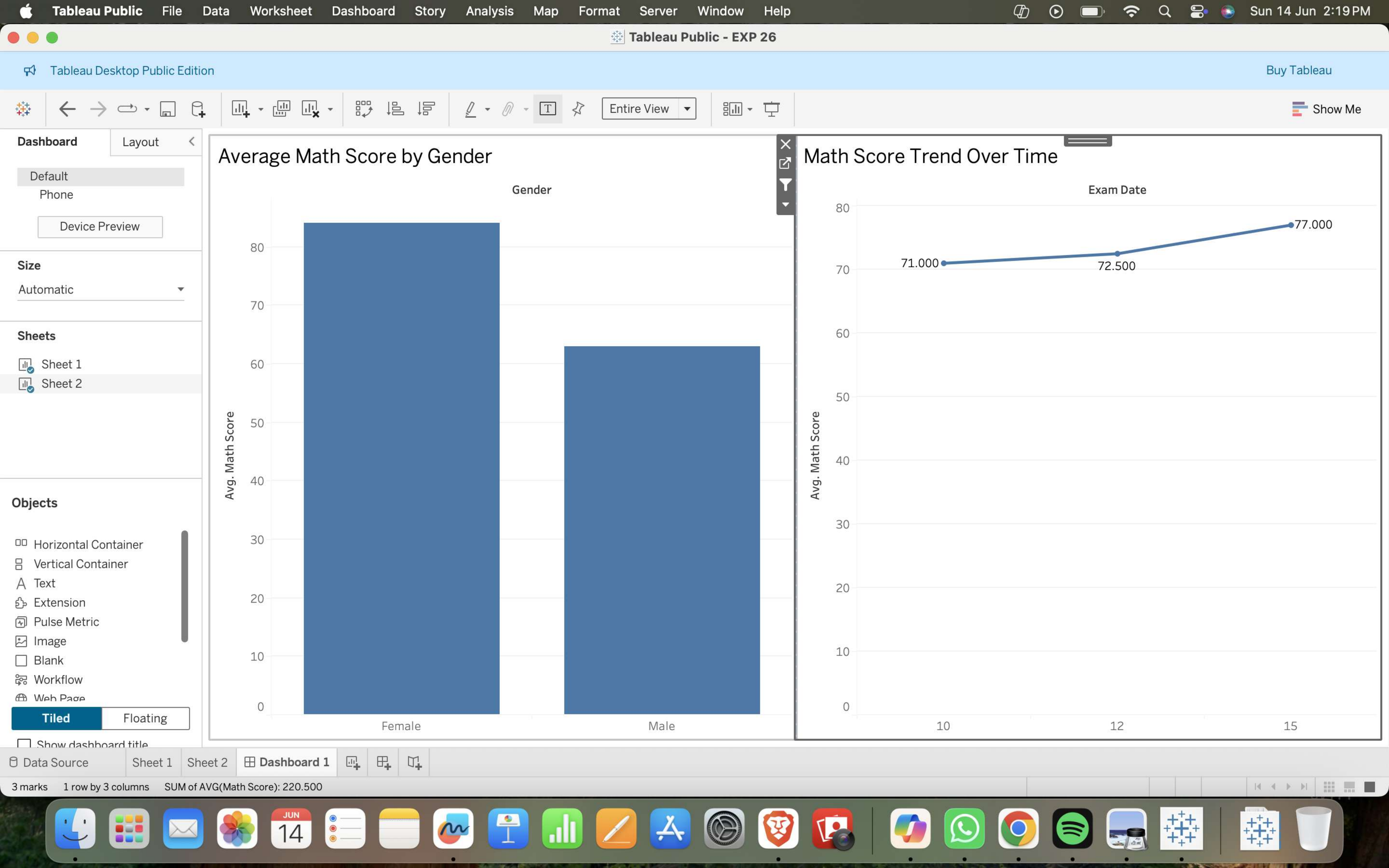
Exam Date
Gender
Student ID
Measure Names
Attendance
Math Score
Science Score
Study Hours
Student_Mini_Data.csv (C...
Measure Values
Pages
Filters
Action (Gender)
Marks
Automatic
Color
Size
Label
Detail
Tooltip
Path
Columns
DAY(Exam Date)
Rows
AVG(Math Score)
Math Score Trend Over Time
Exam Date
Avg. Math Score
77.000
72.500
71.000
80
70
60
50
40
30
20
10
0
10
12
15
Data Source
Sheet 1
Sheet 2
Dashboard 1
3 marks
1 row by 3 columns
SUM of AVG(Math Score): 220.500
JUN
14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Dashboard
Layout
Default
Phone
Device Preview
Size
Automatic
Sheets
Sheet 1
Sheet 2
Objects
Horizontal Container
Vertical Container
Text
Extension
Pulse Metric
Image
Blank
Workflow
Web Page
Tiled
Floating
Show dashboard title
Average Math Score by Gender
Gender
Avg. Math Score
0
10
20
30
40
50
60
70
80
Female
Male
Math Score Trend Over Time
Exam Date
Avg. Math Score
0
10
20
30
40
50
60
70
80
71.000
72.500
77.000
10
12
15
Data Source
Sheet 1
Sheet 2
Dashboard 1
3 marks
1 row by 3 columns
SUM of AVG(Math Score): 220.500
JUN
14